

POWERBOX · REQUIREMENTS

PowerBox requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/powerbox		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

PowerBox requirements

These are first-pass requirements for the concept. Targets are proposals for review, not user-validated needs, and will be checked by calculation at TRL 3 and revised after co-design sessions.

ID	Requirement	Target	Verification (TRL 3 or later)
R1	Store useful energy in one SwapCell pack	400 Wh or more usable at the pack terminals, using 10 to 100 % state of charge	Calculation from SwapCell capacity and the host's discharge cut-off
R2	Run a household through outage evenings	Two evenings of the reference load profile (Table 1, 203 Wh at the loads) on one full pack	Energy budget calculation including conversion losses and standby
R3	Accept four charging sources	(a) DC input 12 to 60 V, up to 200 W, with maximum power point tracking for StepGen and a solar panel; (b) grid through a certified external charger; (c) SunSpoke by swapping in a charged SwapCell pack; (d) SunSpoke's 100 W panel through the DC input	Design review against the input specification table
R4	Recharge quickly from the grid	10 to 100 % in 3 h or less	Charger rating and SwapCell charge curve
R5	Recharge from solar in a day	10 to 100 % in one clear day (4.5 peak sun hours) with a 200 W panel	Solar yield calculation
R6	Use human power efficiently	DC input conversion efficiency 90 % or more from 40 to 100 W, and stable operation (no stall or hunting) across StepGen's voltage and power range	Converter efficiency data; later bench test with StepGen
R7	Provide the outputs households use	2 x USB-C PD (one 100 W, one 60 W), 2 x USB-A (12 W each), 2 x 12 V DC (10 A shared), 1 x AC outlet, 300 W continuous pure sine (THD 3 % or less) with GFCI or RCD; total output limited to 400 W	Design review against the output specification table
R8	Waste little energy when idle	Off: 5 % or less of pack energy per month, including cell self-discharge. Ready (display on, outputs off): 1.0 W or less. Inverter switches itself off after 10 min below 5 W	Standby budget calculation; later measurement
R9	Never back-feed household wiring	No AC input connector that can mate with an AC output; AC output only through a GFCI or RCD outlet; grid charging only through a certified external charger with a DC plug; DC bus 60 V or less	Design review and safety checklist
R10	Be carried by one adult	10 kg (22 lb) or less with pack; 500 x 300 x 280 mm or smaller including handle; one top handle	Massing model, then weighing
R11	Stay within the concept budget	PowerBox parts \$450 or less, excluding the SwapCell pack (budgeted with SwapCell)	Priced BOM
R12	Swap packs and show status simply	Pack swapped by hand in 30 s or less without tools; display shows state of charge, input and output power and estimated time remaining, read from the SwapCell CAN messages	Design review; later timed trial with users

■ Assumptions

Table 1. Reference evening load profile (6 pm to 11 pm outage). Proposed for review; the real profile must come from co-design sessions.

Load	Power	Hours	Energy at the load	Output used
Three LED bulbs	3 x 5 W	5	75 Wh	12 V DC
Wi-Fi router and fibre terminal	12 W	5	60 Wh	12 V DC
Radio	5 W	4	20 Wh	12 V DC
Four phone charges	about 12 Wh each		48 Wh	USB
Total			203 Wh	

- The SwapCell pack is the reference 13S2P pack of the SwapCell interface v0.2: 46.8 V nominal, 39.0 to 54.6 V, about 468 Wh.
- StepGen delivers 40 to 100 W DC; StepGen's own estimate is about 60 to 100 Wh stored per hour of steady stepping.
- Refrigerators, kettles and other high-power or high-surge loads are out of scope (see PBX-PRB-001).
- Budget: the \$450 in `project.yaml` covers the PowerBox itself. A household that needs its own pack also buys a SwapCell pack (about \$370 in prototype parts). Whether the budget should include one pack is proposed, awaiting Amish.